

Chapter 14 — Literature

Finding, screening and citing without fabricating

The highest-demand capability in this room, and the one with the most ways to go quietly wrong.



Start from the failure gallery

By this point in the course you will have seen a fabricated citation from your own field, and what it took to catch it.

- It was fluent. It had plausible authors, a plausible year, and a plausible venue.
- It was caught by resolving the identifier, not by reading the sentence.
- Chapter 5 explained why: the middle band of model knowledge, where it believes it knows.

Everything in this chapter is downstream of that one demonstration.



What survives PDF extraction, and what does not

- Two-column layouts interleave. Text from the left column and the right can arrive as one sentence that neither column contains.
- Equations degrade badly. Sub- and superscripts flatten, symbols are substituted, and the result is often syntactically plausible and numerically wrong.
- Tables lose their structure. A number can migrate a column.

Check the extraction before you reason about the content. A wrong number extracted cleanly looks exactly like a right one.

These failure modes are **extractor-dependent** — they are not properties of PDFs. Name the tool you used when you report them, and re-check when you change it. `TODO(capture)` — run one extraction live on a two-column paper with equations rather than describing it.

This is Chapter 4's discipline applied to a file format: verify the intermediate step before interpreting the result.



Reference managers and the document pipeline

- Keep one source of truth for bibliography data, and export from it. Retyping a reference is how a wrong year enters a paper.
- Choose an export format that survives your document pipeline, and test it on one reference before you have three hundred.
- The identifier — DOI, arXiv ID — is the field that matters. Author and year are convenience; the identifier is the thing you can check.



Screening against stated criteria

- Write the inclusion and exclusion criteria **before** you screen. Same rule as the eval in Chapter 4, and for the same reason.
- Screening is a classification task with a defined answer, which makes it one of the few literature tasks that can be checked.
- **Papers get silently dropped.** A screening pass returns what it returns; it does not report what it failed to consider.

So screen a known subset first — twenty papers where you already know the answer — and measure the false-negative rate before trusting it on the ones you have not read. That is a five-case eval wearing different clothes.



Summarising without laundering

- A summary that drops the authors' stated scope converts a bounded finding into a general claim. That is laundering, and it is usually accidental.
- The scope sentence is the one most often lost: which models, which dataset, which conditions.
- **Keep the hedge.** If the authors wrote "suggests", you do not get to write "shows".

You have seen this failure in this course already: a hardware result, true on a specific benchmark, circulating as though the technology were deployed. Chapter 7 corrected it in front of you.



The habit that closes the thread

Every citation is verified against a primary identifier. No exceptions.

- Resolve the DOI or the arXiv ID. Open the record. Confirm the title, the authors and the year.
- A citation you have not resolved is a candidate, not a reference.
- It costs a few seconds per source once the habit is formed, and it is the entire defence.

Thread — truthfulness. Introduced in Chapter 2 as a rated rubric dimension, explained in Chapter 5 as a structural property rather than a bug, and closed here for citations — the one part of the problem with a mechanical check. The general case does not close: a fluent, uncitable, wrong sentence has no identifier to resolve. That is what Chapter 17 is for.



Exercise

1. Take a paper you know well. Extract it and check what survived — equations first.
2. Ask for a summary. Find the scope sentence in the original and check whether it survived.
3. Ask for five references on a narrow question in your field. **Resolve every identifier**. Record how many were real, how many were real but said something else, and how many did not exist.

Step 3 is the one you will remember. Write the numbers down — they are your own failure gallery, and they are more persuasive to you than anyone else's.

